



**COMSATS University Islamabad
Abbottabad, Pakistan**

HomeEase

**Hyper-Local Domestic Worker Connect Platform with Explainable AI
Recommendation & Bidirectional Marketplace**

By

Hashir Hamid CIIT/FA22-BSE-139/ATD

Aman Ullah Khan CIIT/FA22-BSE-074/ATD

Umar Saeed CIIT/FA22-BCS-041/ATD

Supervisor

Sir Sumair Khan

Bachelor of Science in Computer Science / Software Engineering (2022–2026)

**The candidates confirm that the work submitted is their own and appropriate credit has
been given where reference has been made to the work of others.**



COMSATS University Islamabad, Abbottabad Campus

HomeEase

**A project presented to
COMSATS University Islamabad**

**In partial fulfillment of the requirements for the degree of
Bachelor of Science in Computer Science / Software Engineering
(2022–2026)**

By

Hashir Hamid CIIT/FA22-BSE-139/ATD

Aman Ullah Khan CIIT/FA22-BSE-074/ATD

Umar Saeed CIIT/FA22-BCS-041/ATD

DECLARATION

We hereby declare that this 60% Final Year Project Thesis Report entitled "HomeEase: Hyper-Local Domestic Worker Connect Platform with Explainable AI Recommendation and Bidirectional Marketplace" is our own original work. No portion of this work has been submitted in support of any other application for another degree or qualification at this or any other university or institute of learning. Appropriate credit, citations, and references have been duly provided wherever external concepts, libraries, frameworks, or standards have been consulted.

The project has been developed under the academic supervision of Sir Sumair Khan at the Department of Computer Science, COMSATS University Islamabad, Abbottabad Campus. The active system adheres strictly to the directives and scope refinements issued by the FYP Evaluation Committee on 2026-09-29, introducing an authentic Content-Based AI recommendation engine, a bidirectional job marketplace, bilingual (Urdu/English) localization, and an Abbottabad cold-start evaluation dataset.

Hashir Hamid
FA22-BSE-139

Aman Ullah Khan
FA22-BSE-074

Umar Saeed
FA22-BCS-041

Signature

Signature

Signature

Sir Sumair Khan
Project Supervisor

Signature

EXECUTIVE SUMMARY

In Pakistan, the domestic informal labor sector—encompassing cooks, cleaners, maids, child nannies, and elderly caregivers—remains largely opaque, decentralized, and governed by informal word-of-mouth networks. This informal hiring paradigm introduces severe friction: households confront safety risks, lack of identity verification, and unpredictable service quality, while domestic workers suffer from arbitrary wage negotiation, undefined task scopes, delayed cash compensations, and lack of professional reputation portability. Furthermore, conventional digital platforms fail in the Pakistani local market because they assume universal English literacy, rely on automated credit card payment gateways that domestic workers cannot access, and employ static database dropdown filters disguised as "recommendations".

To overcome these socio-economic and technical deficits, this project presents HomeEase, a hyper-local domestic worker marketplace designed specifically for local Pakistani dynamics, with primary empirical validation in Abbottabad. HomeEase establishes a structured mobile ecosystem built using Flutter and Dart, backed by Supabase PostgreSQL with Row Level Security (RLS). At this 60% evaluation milestone, HomeEase successfully fulfills the four official mandates issued by the FYP Evaluation Committee: (1) an authentic Content-Based Vector Similarity Recommender System combining Cosine Similarity on skill feature spaces, Haversine geospatial proximity decay, and min-max rating normalization with Explainable AI (XAI) transparent match badges; (2) a Bidirectional Job Marketplace allowing households to publish open gigs and domestic workers to actively browse and apply for local neighborhood opportunities; (3) a Bilingual Urdu/English localization layer ('EN | Urdu') with high-affordance visual task icons tailored for low-literacy informal workers; and (4) an Abbottabad synthetic seed dataset and cold-start evaluation benchmark. This document presents the complete 60% system engineering lifecycle, spanning theoretical foundations, rigorous UML design models, relational schema specifications, algorithmic implementations, and comprehensive verification test suites.

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We also thank the faculty members of the Department of Computer Science at COMSATS University Islamabad, Abbottabad Campus, for their rigorous evaluation feedback during the 30% milestone review, which fundamentally elevated the technical depth of this project. Lastly, we owe immense appreciation to our families, fellow students, and our technical mentor AbuZar Babar for their unwavering encouragement, collaborative discussions, and continuous moral support.

ABBREVIATIONS

Abbreviation	Definition
AI	Artificial Intelligence
BaaS	Backend as a Service
CNIC	Computerized National Identity Card (Pakistan NADRA)
ERD	Entity Relationship Diagram
FK	Foreign Key
HCI	Human-Computer Interaction
JWT	JSON Web Token
KYC	Know Your Customer (Identity & Background Verification)
PK	Primary Key
REST	Representational State Transfer
RLS	Row Level Security (PostgreSQL)
SDD	Software Design Description
SRS	Software Requirements Specification
XAI	Explainable Artificial Intelligence

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1. Introduction

1.1 Brief Introduction

HomeEase is an enterprise-grade, hyper-local domestic worker connect platform specifically architected to formalize and secure the informal home-services sector in Pakistan. The platform bridges the deep trust deficit between two primary stakeholders: urban households seeking reliable, safe domestic assistance (such as cooks, cleaners, maids, child nannies, and elderly caregivers), and informal domestic workers who require reliable employment visibility, fair compensation, and professional reputation portability.

In urban and semi-urban Pakistani centers like Abbottabad, domestic worker hiring has historically been restricted to physical word-of-mouth, informal intermediaries, or gatekeeper referrals. These informal channels lack background identity checks, verifiable service histories, transparent task specifications, and standardized wage metrics. Consequently, employers face security risks and unpredictable absenteeism, while domestic workers are exposed to exploitative working hours, arbitrary wage cuts, and zero dispute recourse.

HomeEase establishes a structured digital bridge through a three-tier architecture comprising a Flutter mobile application for both households and workers, a React/Vite admin dashboard for regulatory governance, and a Supabase PostgreSQL backend. At this 60% evaluation stage, HomeEase extends beyond conventional single-sided directory apps by delivering an Explainable AI (XAI) Recommendation Engine, a Bidirectional Job Marketplace, and a specialized Bilingual/Iconographic accessibility interface tailored for low-literacy workers.

1.2 Relevance to Course Modules

The engineering lifecycle and architectural implementation of HomeEase synthesize theoretical knowledge and practical competencies acquired across five core academic modules of the Software Engineering curriculum at COMSATS University Islamabad:

1. CSC392 (Software Engineering): Methodical application of the Agile Scrum methodology, requirement identification techniques, formal SRS and SDD authoring, UML object-oriented modeling, and Software Requirements Traceability Matrices (RTM).
2. CSC341 (Database Systems): Relational database architecture, Third Normal Form (3NF) relational decomposition, entity integrity, foreign key cascading constraints, index optimization, and PostgreSQL Row Level Security (RLS) policies.
3. CSC475 (Artificial Intelligence): Formalization of a Content-Based Recommender System, multi-dimensional feature space vectorization, Cosine Similarity distance metrics in high-dimensional discrete spaces, Haversine spherical distance formulas, min-max normalization, and Explainable AI (XAI) badge synthesis.
4. CSC483 (Mobile Application Development): Cross-platform Flutter architecture, Dart asynchronous programming (Futures, Streams), reactive UI state binding, localized layout

constraints, and native hardware sensor integration.

5. CSC412 (Human-Computer Interaction): Usability engineering, accessibility for low-literacy demographics, bilingual right-to-left (RTL) typography handling, and visual iconography affordance design.

1.3 Project Background

The HomeEase project commenced with the 10% Proposal Milestone, identifying the socio-technical breakdown of informal domestic worker hiring in Pakistan. The subsequent 30% Milestone formulated the complete Software Requirements Specification (SRS v1.0) and Software Design Description (SDD v1.0), along with a 15-screen client-side Flutter prototype executing on an interactive state machine.

During the 60% evaluation review on 2026-09-29, the FYP Evaluation Committee established four mandatory technical directives: (1) implementing an authentic AI recommendation engine rather than simple SQL filter dropdowns; (2) creating a two-sided bidirectional marketplace enabling workers to search and apply for household gigs; (3) introducing bilingual English/Urdu localization with visual iconography for low-literacy workers; and (4) validating the system with an empirical dataset from Abbottabad. This report documents the successful architectural realization of these mandates.

1.4 Literature Review

A rigorous literature review of international and regional on-demand service platforms reveals distinct operational models and limitations:

1. Urban Company (formerly UrbanClap, India/UAE): Employs a full-stack managed marketplace model where the platform sets prices, trains workers, and takes commission. While highly standardized, it requires heavy capital investment, complete digital banking penetration, and excludes informal, low-literacy workers who lack formal trade licenses.
2. Care.com (USA / Global): A subscription-based caregiving directory connecting families with nannies and caregivers. It relies heavily on Social Security Number (SSN) background checks, mandatory monthly credit card subscriptions, and extensive written text profiles, rendering it completely unsuited for the cash-based, low-literacy Pakistani market.
3. Handy & TaskRabbit (USA / Europe): Gig platforms focusing on handyman tasks, cleaning, and furniture assembly. Both rely on automated credit card escrow and gig-worker bidding. Neither accommodates informal cash settlement, nor do they support visual affordance for workers who cannot read Latin scripts.
4. KaamKaaj & Local Classifieds (Pakistan / OLX): Local classified platforms simply publish unverified phone numbers and raw text blurbs. They offer zero identity verification, no auto-generated service agreements, no double-booking prevention, no dispute resolution, and zero recommendation intelligence.

1.5 Analysis from Literature Review

The reviewed platforms were compared across local matching, worker verification, payment accessibility, marketplace direction, recommendation intelligence, language support, and service agreements.

- **Urban Company:** Provides strong worker screening and standardized service terms, but primarily supports card-based payments, platform-assigned work, and text-heavy English/Hindi interfaces.
- **Care.com:** Supports caregiver job postings and collaborative recommendations, but relies on credit-card subscriptions, formal identity systems, English content, and does not generate service agreements.
- **TaskRabbit:** Offers task feeds, third-party identity checks, and digital escrow, but lacks Urdu support, informal payment handling, and interfaces designed for low-literacy domestic workers.
- **Pakistani Classified Platforms:** Support city-level listings and informal cash arrangements, but generally provide no reliable verification, intelligent recommendations, dispute support, payment evidence, or formal agreements.
- **HomeEase:** Combines Abbottabad-focused proximity matching, CNIC and police-document verification, Cash/JazzCash/EasyPaisa receipt tracking, a bidirectional job board, explainable content-based recommendations, bilingual English/Urdu interfaces, visual icons, and auto-generated service agreements.

Critical Research Gap: No reviewed platform combines hyper-local domestic worker connectivity with explainable AI matching, informal payment auditability, formal service agreements, and bilingual accessibility for low-literacy users in semi-urban Pakistan. HomeEase is designed to address this gap.

1.6 Methodology and Software Lifecycle for This Project

HomeEase is developed following the Agile Scrum methodology, organized into two-week development sprints. Agile was selected because domestic worker marketplace workflows require continuous usability testing with both high-literacy employers and low-literacy service providers.

The project lifecycle maps directly to academic milestone checkpoints:

- 10% Milestone (Proposal): Problem definition, literature review, technical feasibility, and high-level module planning.
- 30% Milestone (SRS & SDD): Formal specification of 32 functional requirements, comprehensive UML design suite, 14-table database schema, and client-side Flutter prototype.

- 60% Milestone (Mid-Term Implementation - Active): Incorporation of the four Committee Mandates: Content-Based AI Recommendation Engine with XAI match badges, Bidirectional Job Marketplace, Bilingual Urdu/English Engine with visual icons, Abbottabad synthetic dataset, and Supabase PostgreSQL integration.
- 100% Milestone (Final Defense): Live SMS OTP gateway, comprehensive Usability Testing across Abbottabad demographic cohorts, performance benchmarking, and final thesis.

2. Problem Definition

2.1 Problem Statement

The informal domestic hiring sector in Pakistan suffers from systemic institutional and technological failure, manifested across six core operational pillars:

1. **Complete Lack of Background Verification:** Households admit domestic workers into private residential spaces with zero verifiable identity records. No centralized repository exists to inspect CNIC authenticity, police character clearance, or past disciplinary histories, creating significant personal and physical security risks.
2. **Information Asymmetry and Informal Wage Exploitation:** Absence of standardized rate cards leads to arbitrary price gouging by workers or unfair wage depression by employers. Workers lack reputation portability; five years of honest service for one household cannot be digitally proven when seeking employment with another.
3. **Single-Sided Friction in Gig Discovery:** Traditional platforms treat workers as passive entries in a directory. Workers sitting idle have no mechanism to view nearby households needing immediate assistance, creating severe underemployment.
4. **Exclusionary Literacy and Language Barriers:** The majority of domestic workers in Pakistan cannot read English, and many have limited Urdu text literacy. Digital platforms built exclusively with dense text menus and complex navigation exclude the very demographic they aim to empower.
5. **Cold-Start Failure in Traditional Recommender Systems:** Collaborative filtering algorithms rely on millions of historical interaction matrices, causing complete failure during early deployment. Conversely, simple SQL WHERE queries lack algorithmic intelligence and fail academic standards.
6. **Unrealistic Digital Payment Gateways:** Mandating automated credit card escrow excludes 95% of domestic workers who operate exclusively in cash or basic mobile wallets (EasyPaisa / JazzCash).

2.2 Deliverables and Development Requirements

The HomeEase project delivers a production-style, multi-role domestic-services platform composed of mobile interfaces, an administrative web portal, intelligent matching services, and

secure cloud infrastructure. The main deliverables are:

1. Household Mobile Application (Flutter)

A cross-platform mobile experience for household users to register, manage profiles, discover workers, inspect explainable match scores, publish jobs, request bookings, review agreements, record payments, submit reviews, and raise disputes.

2. Domestic Worker Mobile Application (Flutter)

A role-specific worker interface for profile creation, service and rate management, availability scheduling, verification-document submission, job browsing, application submission, booking responses, payment confirmation, and reputation tracking.

3. Admin Web Dashboard (React, Vite, and TypeScript)

A browser-based governance portal for worker verification, user monitoring, booking and payment audits, dispute handling, evidence review, and basic platform reporting.

4. Content-Based Recommendation Engine

An explainable worker-ranking module combining skill-vector cosine similarity, Haversine geographic proximity, and normalized ratings to generate transparent match scores and recommendation badges.

5. Bidirectional Job Marketplace

A two-sided marketplace where households publish open service opportunities and verified workers browse and apply for suitable jobs based on category and locality.

6. Service Agreement, Payment, and Dispute Modules

Booking-linked agreement generation, manual Cash/JazzCash/EasyPaisa payment evidence, worker payment confirmation, and administrator-mediated dispute resolution.

7. Bilingual and Accessible Interface

English/Urdu localization, right-to-left layout support, and high-affordance service icons designed to improve usability for workers with limited English or text literacy.

8. Supabase Backend and Abbottabad Dataset

A PostgreSQL database with authentication, Row Level Security, object storage, and a representative Abbottabad seed dataset for development and cold-start recommendation evaluation.

Table 2.1 HomeEase Development Requirements

Category	Requirement	Purpose in HomeEase
Mobile Frontend	Flutter SDK 3.x and Dart 3.x	Build household and domestic-worker mobile interfaces for Android and future iOS deployment.
Admin Frontend	React, Vite, TypeScript, and Tailwind CSS	Build the browser-based administration, verification, audit, and dispute-management portal.
Backend Platform	Supabase cloud services	Provide managed authentication, database APIs, storage, and real-time data synchronization.
Database	PostgreSQL 15+	Store relational records for users, profiles, jobs, bookings, agreements, payments, reviews, and disputes.
Authentication	Supabase Auth and JWT sessions	Secure account access and enforce household, worker, and administrator roles.
Security	PostgreSQL Row Level Security policies	Restrict sensitive profiles, CNIC records, receipts, and dispute evidence to authorized users.
File Storage	Supabase Storage buckets	Store verification documents, payment receipts, profile images, and optional dispute evidence.
Recommendation Logic	Dart/Python vector scoring	Compute explainable skill, location, and rating-based worker recommendations.
Localization	English/Urdu resource dictionaries	Support bilingual content, Urdu rendering, and right-to-left interface behavior.
Development Tools	Android Studio, Visual Studio Code, Git, and GitHub	Support implementation, testing, collaboration, and version control.
Design Tools	Figma and Draw.io/StarUML	Produce user-interface mockups, architecture diagrams, and UML design models.

3. Requirement Analysis

3.1 System Boundary & Use Case Diagram

The HomeEase system boundary encompasses three primary external actors (Household User, Domestic Worker User, and Platform Administrator) interacting with core backend services (Authentication Service, Database Service, AI Recommendation Service, and Notification Service).

Household users execute worker discovery, inspect AI match badges, post open gigs, initiate bookings, sign formal agreements, log manual payments, and file disputes. Domestic workers manage digital profiles, submit KYC documents, set availability calendars, browse open household gigs, submit job proposals, accept bookings, and confirm payment receipts. Administrators govern verification queues, audit transactions, and mediate disputes.

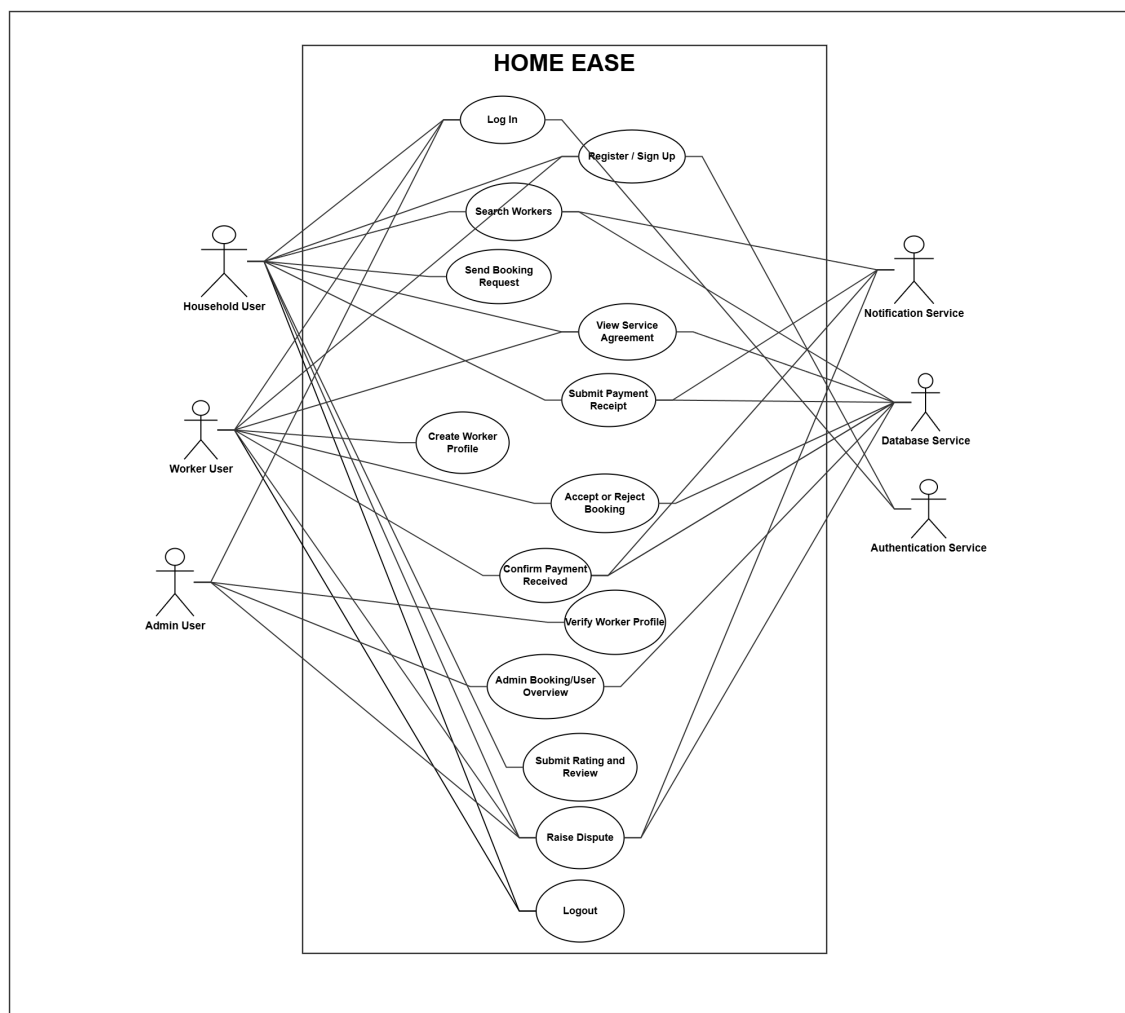


Figure 3.1 Complete Use Case Diagram for the HomeEase Platform

3.2 Detailed Use Cases

The following use case specifications detail the primary interactions across household, worker, administrative, and algorithmic system actors:

3.2.1 Household User Use Cases

UC-HH-01: Household Registration and Profile Setup

Table 3.1 UC-HH-01: Household Registration and Profile Setup

Specification Field	Detailed System Behavior
Use Case ID	UC-HH-01
Use Case Name	Household Registration & Profile Setup
Primary Actor	Household User
Description	A household employer registers a secure account and configures neighborhood location preferences.
Trigger Event	User selects 'Register as Household' on the mobile onboarding screen.
Preconditions	Device has network connectivity. User has a valid mobile number or email.
Postconditions	User account and household profile record are created in Supabase database.
Main Success Flow	1. User opens app and selects Household role. 2. User enters full name, email/phone, password, and Abbottabad locality (e.g., Mandian). 3. System validates credential formats. 4. System creates authentication record and household profile. 5. User is redirected to Household Discovery Home.

UC-HH-02: Discover Workers via Content-Based AI Recommender

Table 3.2 UC-HH-02: Discover Workers via Content-Based AI Recommender

Specification Field	Detailed System Behavior
Use Case ID	UC-HH-02
Use Case Name	Discover Workers via Content-Based AI Recommender
Primary Actor	Household User
Description	Household views an algorithmically ranked list of domestic workers tailored to service category, geographic proximity, and ratings.
Trigger Event	User opens the HomeEase search hub and selects a service category.
Preconditions	Household location is set. Worker profiles exist in the target region.
Postconditions	System displays sorted worker cards accompanied by Explainable AI (XAI) match percentage badges.
Main Success Flow	1. Household selects service category (e.g., Cooking). 2. System extracts household feature vector (category, locality coordinates). 3. AI Recommendation Engine computes Cosine similarity, Haversine distance decay, and normalized ratings across active worker profiles. 4. System ranks workers by composite score. 5. UI displays worker cards with match percentage (e.g., '94% Match') and reasoning tags.

UC-HH-03: Publish Open Household Gig or Job Post

Table 3.3 UC-HH-03: Publish Open Household Gig or Job Post

Specification Field	Detailed System Behavior
Use Case ID	UC-HH-03
Use Case Name	Publish Open Household Gig / Job Post
Primary Actor	Household User
Description	Household publishes an open domestic task request for local workers to browse and apply (Bidirectional Marketplace).
Trigger Event	Household selects 'Post a Job' from the mobile dashboard.
Preconditions	Household user is authenticated.
Postconditions	New record is created in JOB_POSTS with status 'open' and dispatched to worker feeds.
Main Success Flow	1. Household enters task title, service category, date/time, Abbottabad locality, and proposed budget. 2. User submits the gig posting. 3. System validates inputs and persists job record. 4. Notification service alerts matching domestic workers in the vicinity. 5. Job appears in the Worker 'Available Jobs' feed.

UC-HH-04: Direct Booking Request Initiation

Table 3.4 UC-HH-04: Direct Booking Request Initiation

Specification Field	Detailed System Behavior
Use Case ID	UC-HH-04
Use Case Name	Direct Booking Request Initiation
Primary Actor	Household User
Description	Household books a specific domestic worker for a defined date, time, and service scope.
Trigger Event	Household taps 'Book Now' on a worker profile screen.
Preconditions	Worker is verified and has open availability slots for the requested date.
Postconditions	Booking record created with status 'Pending'. Double-booking lock engaged.
Main Success Flow	1. Household selects date, start time, end time, and service tasks. 2. System checks worker availability and verifies no scheduling conflict. 3. Household reviews booking summary and taps 'Send Booking Request'. 4. System creates booking record in BOOKINGS. 5. Notification sent to worker's mobile device.

UC-HH-05: Sign Auto-Generated Formal Service Agreement

Table 3.5 UC-HH-05: Sign Auto-Generated Formal Service Agreement

Specification Field	Detailed System Behavior
Use Case ID	UC-HH-05
Use Case Name	Sign Auto-Generated Formal Service Agreement
Primary Actor	Household & Worker
Description	Both parties inspect and digitally sign an auto-generated service contract prior to service commencement.
Trigger Event	Worker accepts a pending booking request.
Preconditions	Booking status is 'Accepted'.
Postconditions	Service agreement record generated in SERVICE_AGREEMENTS with status 'Signed'.
Main Success Flow	1. System compiles agreed task items, working hours, total wage, locality address, and dispute cancellation policy. 2. Household reviews contract text and clicks 'Agree & Sign'. 3. Worker reviews contract on their mobile device and clicks 'Confirm Agreement'. 4. Agreement status transitions to 'Active'.

UC-HH-06: Log Manual Payment and Upload Receipt

Table 3.6 UC-HH-06: Log Manual Payment and Upload Receipt

Specification Field	Detailed System Behavior
Use Case ID	UC-HH-06
Use Case Name	Log Manual Payment & Upload Receipt
Primary Actor	Household User
Description	Household logs cash or mobile wallet payment (JazzCash/EasyPaisha) and uploads receipt proof.
Trigger Event	Service agreement reaches completion date.
Preconditions	Booking status is 'Completed'.
Postconditions	Record created in PAYMENT_RECORDS with status 'Submitted'. Worker notified to confirm.
Main Success Flow	1. Household selects payment method (Cash, JazzCash, EasyPaisha). 2. Household enters transaction ID / reference and amount paid. 3. User captures image of receipt / screenshot and uploads. 4. System uploads image to Supabase Storage bucket. 5. Notification dispatched to Worker for confirmation.

3.2.2 Domestic Worker Use Cases

UC-WK-03: Browse Available Jobs Feed

Table 3.7 UC-WK-03: Browse Available Jobs Feed

Specification Field	Detailed System Behavior
Use Case ID	UC-WK-03
Use Case Name	Browse Available Jobs Feed (Bidirectional Marketplace)
Primary Actor	Domestic Worker
Description	Worker browses open household job postings matching their trade and location.
Trigger Event	Worker taps 'Find Jobs' tab on the mobile dashboard.
Preconditions	Worker profile has active service categories assigned.
Postconditions	Worker views scrollable list of open gigs with budget, locality, and time requirements.
Main Success Flow	1. Worker opens 'Available Jobs' feed. 2. System queries JOB_POSTS where status = 'open' matching worker category and locality. 3. Worker inspects job requirements and budget. 4. Worker taps 'Apply Now'. 5. System registers application in JOB_APPLICATIONS and alerts household.

3.2.3 System and General Use Cases

UC-SYS-01: Content-Based AI Worker Scoring Engine

Table 3.8 UC-SYS-01: Content-Based AI Worker Scoring Engine

Specification Field	Detailed System Behavior
Use Case ID	UC-SYS-01
Use Case Name	Content-Based AI Worker Scoring Engine
Primary Actor	System Recommender
Description	Calculates multi-dimensional vector similarity and geographic decay for worker recommendations.
Trigger Event	Household initiates worker search or opens discovery hub.
Preconditions	Household location coordinates and target category vector available.
Postconditions	Normalized match scores and Explainable AI badge strings generated for all candidate workers.
Main Success Flow	1. System extracts binary category/skill vector u for household requirement. 2. System computes Cosine Similarity against worker vectors w_i. 3. System computes Haversine distance d_i in kilometers between household and worker coordinates. 4. System computes proximity decay score $S_{geo} = 1 / (1 + 0.2 d_i)$. 5. System normalizes worker ratings to $[0, 1]$. 6. System calculates composite score $S = 0.50 S_{skills} + 0.35 S_{geo} + 0.15 S_{rating}$. 7. System formats explainability string (e.g., '94% Match • 1.2 km away • Desi Cooking fit').

UC-GEN-01: Bilingual Language and Visual Affordance Toggle

Table 3.9 UC-GEN-01: Bilingual Language and Visual Affordance Toggle

Specification Field	Detailed System Behavior
Use Case ID	UC-GEN-01
Use Case Name	Bilingual Language & Visual Affordance Toggle
Primary Actor	Household & Worker
Description	User toggles application language between English and Urdu ('Urdu') with visual icon adaptation.
Trigger Event	User taps the language toggle button ('EN Urdu') in the app bar.
Preconditions	Application is running on device.
Postconditions	All screen strings re-render in selected language; RTL direction enabled for Urdu.
Main Success Flow	1. User clicks language toggle. 2. Localization manager updates active locale state. 3. String resources switch between English and Urdu dictionaries. 4. Layout direction mirrors to Right-To-Left (RTL) for Urdu. 5. Service category cards display high-affordance pictorial icons (Broom, Pot, Stroller, Wheelchair) for immediate visual recognition by low-literacy workers.

3.3 Functional Requirements

The HomeEase functional requirements are grouped by subsystem. Each requirement is documented separately using the same specification format as the TrekPal reference.

3.3.1 Authentication Functional Requirements

Authentication and role-management requirements are as follows.

FR-01: User Account Registration

Table 3.10 FR-01: User Account Registration

Field	Description
Identifier	FR-01
Title	User Account Registration
Requirement	The system shall allow household and worker users to register accounts using email/phone and secure password.
Source	Authentication and role-management workflow
Rationale	Required to provide secure, role-appropriate access to HomeEase.
Business Rules	Credentials must be valid and each account must have one authorized role.
Dependencies	Supabase Auth, user profile records, role-based navigation
Priority	High

FR-02: Role-Based User Authentication

Table 3.11 FR-02: Role-Based User Authentication

Field	Description
Identifier	FR-02
Title	Role-Based User Authentication
Requirement	The system shall authenticate registered users and issue role-specific JSON Web Tokens (JWT).
Source	Authentication and role-management workflow
Rationale	Required to provide secure, role-appropriate access to HomeEase.
Business Rules	Credentials must be valid and each account must have one authorized role.
Dependencies	Supabase Auth, user profile records, role-based navigation
Priority	High

FR-03: Role Selection and Demo Login

Table 3.12 FR-03: Role Selection and Demo Login

Field	Description
Identifier	FR-03
Title	Role Selection and Demo Login
Requirement	The mobile login interface shall allow instant switching and demo credential auto-fill for testing.
Source	Authentication and role-management workflow
Rationale	Required to provide secure, role-appropriate access to HomeEase.
Business Rules	Credentials must be valid and each account must have one authorized role.
Dependencies	Supabase Auth, user profile records, role-based navigation
Priority	Medium

FR-04: Role-Based Route Protection

Table 3.13 FR-04: Role-Based Route Protection

Field	Description
Identifier	FR-04
Title	Role-Based Route Protection
Requirement	The system shall enforce strict role-based route guards preventing unauthorized cross-role access.
Source	Authentication and role-management workflow
Rationale	Required to provide secure, role-appropriate access to HomeEase.
Business Rules	Credentials must be valid and each account must have one authorized role.
Dependencies	Supabase Auth, user profile records, role-based navigation
Priority	High

3.3.2 Household, Worker, and Verification Functional Requirements

Profile management, worker services, availability, and verification requirements are as follows.

FR-05: Household Profile Management

Table 3.14 FR-05: Household Profile Management

Field	Description
Identifier	FR-05
Title	Household Profile Management
Requirement	The system shall allow households to manage profile details, contact numbers, and Abbottabad locality address.
Source	UC-HH-01 and household profile workflow
Rationale	Required to store household identity, contact, and service-location information.
Business Rules	Only the authenticated household may update its profile.
Dependencies	Authentication service, household profile table
Priority	High

FR-06: Worker Professional Profile Management

Table 3.15 FR-06: Worker Professional Profile Management

Field	Description
Identifier	FR-06
Title	Worker Professional Profile Management
Requirement	The system shall allow domestic workers to create professional profiles with bio, experience years, and locality.
Source	Worker profile and service-management workflow
Rationale	Required to present accurate worker skills, pricing, experience, and availability.
Business Rules	Only the profile owner or an authorized administrator may modify worker data.
Dependencies	Worker profile, service category, pricing, and availability records
Priority	High

FR-07: Worker Service Category Selection

Table 3.16 FR-07: Worker Service Category Selection

Field	Description
Identifier	FR-07
Title	Worker Service Category Selection
Requirement	The system shall allow workers to select multiple service categories (Cooking, Cleaning, Childcare, Elderly Care).
Source	Worker profile and service-management workflow
Rationale	Required to present accurate worker skills, pricing, experience, and availability.
Business Rules	Only the profile owner or an authorized administrator may modify worker data.
Dependencies	Worker profile, service category, pricing, and availability records
Priority	High

FR-08: Worker Service Rate Configuration

Table 3.17 FR-08: Worker Service Rate Configuration

Field	Description
Identifier	FR-08
Title	Worker Service Rate Configuration
Requirement	The system shall allow workers to configure hourly or visit-based charge rates per service category.
Source	Worker profile and service-management workflow
Rationale	Required to present accurate worker skills, pricing, experience, and availability.
Business Rules	Only the profile owner or an authorized administrator may modify worker data.
Dependencies	Worker profile, service category, pricing, and availability records
Priority	Medium

FR-09: Worker Availability Scheduling

Table 3.18 FR-09: Worker Availability Scheduling

Field	Description
Identifier	FR-09
Title	Worker Availability Scheduling
Requirement	The system shall provide a weekly calendar allowing workers to toggle availability time slots by day of week.
Source	Worker profile and service-management workflow
Rationale	Required to present accurate worker skills, pricing, experience, and availability.
Business Rules	Only the profile owner or an authorized administrator may modify worker data.
Dependencies	Worker profile, service category, pricing, and availability records
Priority	High

FR-10: Worker Verification Document Submission

Table 3.19 FR-10: Worker Verification Document Submission

Field	Description
Identifier	FR-10
Title	Worker Verification Document Submission
Requirement	The system shall allow workers to upload CNIC front/back images and character certificates for verification.
Source	Worker verification workflow
Rationale	Required to establish worker identity and platform trust.
Business Rules	Required documents must be complete before administrative review.
Dependencies	Worker profile, secure storage, admin verification queue
Priority	High

3.3.3 Recommendation and Job Marketplace Functional Requirements

AI matching and bidirectional job-marketplace requirements are as follows.

FR-11: Skill Vector Generation

Table 3.20 FR-11: Skill Vector Generation

Field	Description
Identifier	FR-11
Title	Skill Vector Generation
Requirement	The system shall vectorize household requirements and compute Content-Based Cosine Similarity on worker skills.
Source	UC-SYS-01 Content-Based AI Worker Scoring Engine
Rationale	Required to rank suitable workers transparently for household users.
Business Rules	Scores must use available skill, location, and rating data without hidden manual ordering.
Dependencies	Worker services, household requirements, coordinates, ratings
Priority	High

FR-12: Geographic Distance Calculation

Table 3.21 FR-12: Geographic Distance Calculation

Field	Description
Identifier	FR-12
Title	Geographic Distance Calculation
Requirement	The system shall calculate Haversine geographic distance between household and worker coordinates.
Source	UC-SYS-01 Content-Based AI Worker Scoring Engine
Rationale	Required to rank suitable workers transparently for household users.
Business Rules	Scores must use available skill, location, and rating data without hidden manual ordering.
Dependencies	Worker services, household requirements, coordinates, ratings
Priority	High

FR-13: Explainable Composite Match Scoring

Table 3.22 FR-13: Explainable Composite Match Scoring

Field	Description
Identifier	FR-13
Title	Explainable Composite Match Scoring
Requirement	The system shall compute a composite recommendation match score and display transparent Explainable AI badges.
Source	UC-SYS-01 Content-Based AI Worker Scoring Engine
Rationale	Required to rank suitable workers transparently for household users.
Business Rules	Scores must use available skill, location, and rating data without hidden manual ordering.
Dependencies	Worker services, household requirements, coordinates, ratings
Priority	High

FR-14: Household Job Post Creation

Table 3.23 FR-14: Household Job Post Creation

Field	Description
Identifier	FR-14
Title	Household Job Post Creation
Requirement	The system shall allow households to publish open domestic job postings with category, budget, date, and locality.
Source	UC-HH-03 and UC-WK-03
Rationale	Required to support a bidirectional marketplace where households post jobs and workers apply.
Business Rules	Only open jobs may appear in worker feeds or accept applications.
Dependencies	Job posts, job applications, profiles, notification service
Priority	High

FR-15: Worker Job Feed

Table 3.24 FR-15: Worker Job Feed

Field	Description
Identifier	FR-15
Title	Worker Job Feed
Requirement	The system shall display a live feed of open household gigs to domestic workers matching their category and locality.
Source	UC-HH-03 and UC-WK-03
Rationale	Required to support a bidirectional marketplace where households post jobs and workers apply.
Business Rules	Only open jobs may appear in worker feeds or accept applications.
Dependencies	Job posts, job applications, profiles, notification service
Priority	High

FR-16: Worker Job Application

Table 3.25 FR-16: Worker Job Application

Field	Description
Identifier	FR-16
Title	Worker Job Application
Requirement	The system shall allow domestic workers to apply to open household job postings with a single tap.
Source	UC-HH-03 and UC-WK-03
Rationale	Required to support a bidirectional marketplace where households post jobs and workers apply.
Business Rules	Only open jobs may appear in worker feeds or accept applications.
Dependencies	Job posts, job applications, profiles, notification service
Priority	High

FR-17: Household Job Application Notification

Table 3.26 FR-17: Household Job Application Notification

Field	Description
Identifier	FR-17
Title	Household Job Application Notification
Requirement	The system shall notify households immediately upon receiving worker job applications.
Source	UC-HH-03 and UC-WK-03
Rationale	Required to support a bidirectional marketplace where households post jobs and workers apply.
Business Rules	Only open jobs may appear in worker feeds or accept applications.
Dependencies	Job posts, job applications, profiles, notification service
Priority	High

3.3.4 Booking and Service Agreement Functional Requirements

Booking lifecycle and service-agreement requirements are as follows.

FR-18: Direct Booking Request

Table 3.27 FR-18: Direct Booking Request

Field	Description
Identifier	FR-18
Title	Direct Booking Request
Requirement	The system shall allow households to create direct booking requests with date, start time, end time, and notes.
Source	UC-HH-04 Direct Booking Request Initiation
Rationale	Required to coordinate service requests and maintain a valid booking lifecycle.
Business Rules	Bookings must reference valid users, services, dates, and non-overlapping worker times.
Dependencies	Household profile, worker profile, availability, notification service
Priority	High

FR-19: Double-Booking Prevention

Table 3.28 FR-19: Double-Booking Prevention

Field	Description
Identifier	FR-19
Title	Double-Booking Prevention
Requirement	The system shall prevent double-booking by verifying worker availability and detecting overlapping booking intervals.
Source	UC-HH-04 Direct Booking Request Initiation
Rationale	Required to coordinate service requests and maintain a valid booking lifecycle.
Business Rules	Bookings must reference valid users, services, dates, and non-overlapping worker times.
Dependencies	Household profile, worker profile, availability, notification service
Priority	High

FR-20: Worker Booking Response

Table 3.29 FR-20: Worker Booking Response

Field	Description
Identifier	FR-20
Title	Worker Booking Response
Requirement	The system shall allow workers to accept or reject incoming booking requests with immediate household alert.
Source	UC-HH-04 Direct Booking Request Initiation
Rationale	Required to coordinate service requests and maintain a valid booking lifecycle.
Business Rules	Bookings must reference valid users, services, dates, and non-overlapping worker times.
Dependencies	Household profile, worker profile, availability, notification service
Priority	High

FR-21: Booking Status Management

Table 3.30 FR-21: Booking Status Management

Field	Description
Identifier	FR-21
Title	Booking Status Management
Requirement	The system shall maintain booking state transitions across: Pending, Accepted, Rejected, Completed, and Cancelled.
Source	UC-HH-04 Direct Booking Request Initiation
Rationale	Required to coordinate service requests and maintain a valid booking lifecycle.
Business Rules	Bookings must reference valid users, services, dates, and non-overlapping worker times.
Dependencies	Household profile, worker profile, availability, notification service
Priority	High

FR-22: Automatic Service Agreement Generation

Table 3.31 FR-22: Automatic Service Agreement Generation

Field	Description
Identifier	FR-22
Title	Automatic Service Agreement Generation
Requirement	The system shall auto-generate a structured service agreement specifying tasks, hours, rate, and terms upon acceptance.
Source	UC-HH-05 Service Agreement workflow
Rationale	Required to document the accepted scope, timing, and payment terms.
Business Rules	An agreement may be generated only for an accepted booking.
Dependencies	Accepted booking, household and worker confirmation
Priority	High

FR-23: Digital Agreement Confirmation

Table 3.32 FR-23: Digital Agreement Confirmation

Field	Description
Identifier	FR-23
Title	Digital Agreement Confirmation
Requirement	The system shall require both household and worker to view and digitally sign the agreement before job start.
Source	UC-HH-05 Service Agreement workflow
Rationale	Required to document the accepted scope, timing, and payment terms.
Business Rules	An agreement may be generated only for an accepted booking.
Dependencies	Accepted booking, household and worker confirmation
Priority	High

3.3.5 Payment, Review, and Dispute Functional Requirements

Payment tracking, reviews, and dispute-management requirements are as follows.

FR-24: Manual Payment Recording

Table 3.33 FR-24: Manual Payment Recording

Field	Description
Identifier	FR-24
Title	Manual Payment Recording
Requirement	The system shall allow households to log manual payments made via Cash, JazzCash, or EasyPaisa.
Source	UC-HH-06 Manual Payment and Receipt workflow
Rationale	Required to maintain auditable evidence for cash and mobile-wallet payments.
Business Rules	Payment records must remain linked to the correct completed booking.
Dependencies	Booking, service agreement, storage bucket, notification service
Priority	High

FR-25: Payment Receipt Upload

Table 3.34 FR-25: Payment Receipt Upload

Field	Description
Identifier	FR-25
Title	Payment Receipt Upload
Requirement	The system shall allow households to upload digital receipt images or transaction screenshots.
Source	UC-HH-06 Manual Payment and Receipt workflow
Rationale	Required to maintain auditable evidence for cash and mobile-wallet payments.
Business Rules	Payment records must remain linked to the correct completed booking.
Dependencies	Booking, service agreement, storage bucket, notification service
Priority	High

FR-26: Worker Payment Confirmation

Table 3.35 FR-26: Worker Payment Confirmation

Field	Description
Identifier	FR-26
Title	Worker Payment Confirmation
Requirement	The system shall allow workers to inspect payment details and confirm payment receipt.
Source	UC-HH-06 Manual Payment and Receipt workflow
Rationale	Required to maintain auditable evidence for cash and mobile-wallet payments.
Business Rules	Payment records must remain linked to the correct completed booking.
Dependencies	Booking, service agreement, storage bucket, notification service
Priority	High

FR-27: Household Rating and Review

Table 3.36 FR-27: Household Rating and Review

Field	Description
Identifier	FR-27
Title	Household Rating and Review
Requirement	The system shall allow households to submit segmented ratings for punctuality, behavior, and quality (1-5 stars).
Source	Post-service rating and review workflow
Rationale	Required to build worker reputation and support informed hiring decisions.
Business Rules	A household may review only a completed booking and may submit one review per booking.
Dependencies	Completed booking, household profile, worker rating data
Priority	High

FR-28: Worker Rating Recalculation

Table 3.37 FR-28: Worker Rating Recalculation

Field	Description
Identifier	FR-28
Title	Worker Rating Recalculation
Requirement	The system shall automatically recompute worker cumulative average ratings upon review submission.
Source	Post-service rating and review workflow
Rationale	Required to build worker reputation and support informed hiring decisions.
Business Rules	A household may review only a completed booking and may submit one review per booking.
Dependencies	Completed booking, household profile, worker rating data
Priority	High

FR-29: Booking or Payment Dispute Creation

Table 3.38 FR-29: Booking or Payment Dispute Creation

Field	Description
Identifier	FR-29
Title	Booking or Payment Dispute Creation
Requirement	The system shall allow either party to raise a transaction dispute linked to a specific booking or payment record.
Source	Dispute-resolution workflow
Rationale	Required to provide a documented process for booking, service, and payment complaints.
Business Rules	Disputes must identify the related transaction and preserve submitted evidence.
Dependencies	Booking or payment record, evidence storage, admin review
Priority	Medium

FR-30: Dispute Evidence Submission

Table 3.39 FR-30: Dispute Evidence Submission

Field	Description
Identifier	FR-30
Title	Dispute Evidence Submission
Requirement	The system shall allow users to upload evidence photos and descriptions for admin dispute mediation.
Source	Dispute-resolution workflow
Rationale	Required to provide a documented process for booking, service, and payment complaints.
Business Rules	Disputes must identify the related transaction and preserve submitted evidence.
Dependencies	Booking or payment record, evidence storage, admin review
Priority	Medium

3.3.6 Administration, Localization, and Accessibility Functional Requirements

Administrative governance, localization, and accessibility requirements are as follows.

FR-31: Secure Admin Dashboard

Table 3.40 FR-31: Secure Admin Dashboard

Field	Description
Identifier	FR-31
Title	Secure Admin Dashboard
Requirement	The system shall provide a secure React web dashboard for platform administrators.
Source	Administrator governance requirements
Rationale	Required to verify workers, monitor activity, and resolve platform issues.
Business Rules	Only authorized administrators may access governance actions and sensitive evidence.
Dependencies	Admin authentication, platform records, audit permissions
Priority	High

FR-32: Worker Verification Queue

Table 3.41 FR-32: Worker Verification Queue

Field	Description
Identifier	FR-32
Title	Worker Verification Queue
Requirement	The admin web dashboard shall display a worker verification queue with full-resolution document inspection.
Source	Administrator governance requirements
Rationale	Required to verify workers, monitor activity, and resolve platform issues.
Business Rules	Only authorized administrators may access governance actions and sensitive evidence.
Dependencies	Admin authentication, platform records, audit permissions
Priority	High

FR-33: Administrative Dispute Resolution

Table 3.42 FR-33: Administrative Dispute Resolution

Field	Description
Identifier	FR-33
Title	Administrative Dispute Resolution
Requirement	The admin web dashboard shall provide dispute mediation tools allowing admins to review evidence and resolve claims.
Source	Administrator governance requirements
Rationale	Required to verify workers, monitor activity, and resolve platform issues.
Business Rules	Only authorized administrators may access governance actions and sensitive evidence.
Dependencies	Admin authentication, platform records, audit permissions
Priority	High

FR-34: Administrative Audit and CSV Export

Table 3.43 FR-34: Administrative Audit and CSV Export

Field	Description
Identifier	FR-34
Title	Administrative Audit and CSV Export
Requirement	The admin web dashboard shall provide platform audit tables for users, bookings, and payments with CSV export.
Source	Administrator governance requirements
Rationale	Required to verify workers, monitor activity, and resolve platform issues.
Business Rules	Only authorized administrators may access governance actions and sensitive evidence.
Dependencies	Admin authentication, platform records, audit permissions
Priority	Medium

FR-35: Bilingual English and Urdu Interface

Table 3.44 FR-35: Bilingual English and Urdu Interface

Field	Description
Identifier	FR-35
Title	Bilingual English and Urdu Interface
Requirement	The mobile app shall provide a global bilingual toggle supporting English and Urdu ('Urdu') with RTL layout.
Source	UC-GEN-01 Bilingual Language Toggle
Rationale	Required to make the application usable for English- and Urdu-speaking users.
Business Rules	The selected language must update visible labels and layout direction consistently.
Dependencies	Localization resources, locale state, RTL layout support
Priority	High

FR-36: Low-Literacy Visual Accessibility

Table 3.45 FR-36: Low-Literacy Visual Accessibility

Field	Description
Identifier	FR-36
Title	Low-Literacy Visual Accessibility
Requirement	The system shall render high-affordance pictorial icons for all service tasks and booking states to support low-literacy workers.
Source	UC-GEN-01 Visual Affordance requirements
Rationale	Required to support domestic workers with limited text literacy.
Business Rules	Critical actions must include recognizable icons and clear visual status cues.
Dependencies	UI component library, localized labels, accessibility assets
Priority	High

3.4 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints that HomeEase must satisfy.

3.4.1 Usability Requirements

- **NFR-USE-01 - Simple and Accessible Navigation**

All critical worker interactions, including accepting bookings, browsing jobs, and confirming payments, shall be executable within a maximum of three taps from the home screen.

- **NFR-USE-02 - Low-Literacy Visual Interface**

Service categories and primary actions shall use clear, high-contrast pictorial icons that users can recognize without relying entirely on written text.

- **NFR-USE-03 - Bilingual Interface**

The application shall support English and Urdu, with readable Urdu typography, correct right-to-left alignment, and consistent spacing throughout the interface.

3.4.2 Performance Requirements

- **NFR-PERF-01 - Recommendation Response Time**

The Content-Based AI recommendation engine shall calculate match scores for up to 500 active worker profiles within 250 milliseconds under normal operating conditions.

- **NFR-PERF-02 - Interface Responsiveness**

Mobile interface transitions and scrolling shall target 60 frames per second on supported devices.

3.4.3 Reliability Requirements

- **NFR-REL-01 - Scheduling Integrity**

The system shall prevent overlapping accepted bookings for the same worker and time period.

- **NFR-REL-02 - Service Availability**

The platform shall maintain at least 99.5% service availability, excluding announced maintenance periods.

3.4.4 Security Requirements

- **NFR-SEC-01 - Secure Authentication**

User credentials shall be managed securely through Supabase Authentication and shall never be stored as plain text in the application database.

- **NFR-SEC-02 - Protected Documents**

CNIC scans, character certificates, and payment receipts stored in Supabase Storage shall

be protected by access policies that permit only the document owner and authorized administrators.

- **NFR-SEC-03 - Role-Based Access Control**

Households, workers, and administrators shall only access operations and records permitted for their assigned roles.

3.4.5 Portability and Maintainability Requirements

- **NFR-PORT-01 - Cross-Platform Compatibility**

The Flutter mobile application shall operate consistently on Android API level 24 or later and iOS 13 or later.

- **NFR-MNT-01 - Modular Architecture**

Business logic, data access, and presentation components shall remain separated through reusable services and repositories to support testing and future maintenance.

4. Design and Architecture

4.1 System Architecture

HomeEase is designed around a three-tier, service-oriented architecture incorporating modern cloud infrastructure and local algorithmic intelligence. The architectural decomposition comprises:

1. **Presentation Tier:** Cross-platform Flutter mobile client providing role-adaptive interfaces for households and workers, integrated with a dynamic Bilingual Engine (EN/Urdu) and a Visual Iconography Affordance layer. Platform administrators access a dedicated React + Vite + TypeScript web dashboard.
2. **Application & Service Tier:** Decoupled business logic services governing Authentication, Booking state machines, Service Agreement generation, Bidirectional Job Marketplace dispatching, Dispute tracking, and the Content-Based AI Recommendation Engine.
3. **Data & Cloud Storage Tier:** Supabase managed PostgreSQL 15+ relational database with Row Level Security (RLS) policy enforcement, PostgREST API layer, and encrypted object storage buckets for KYC documents and payment receipts.

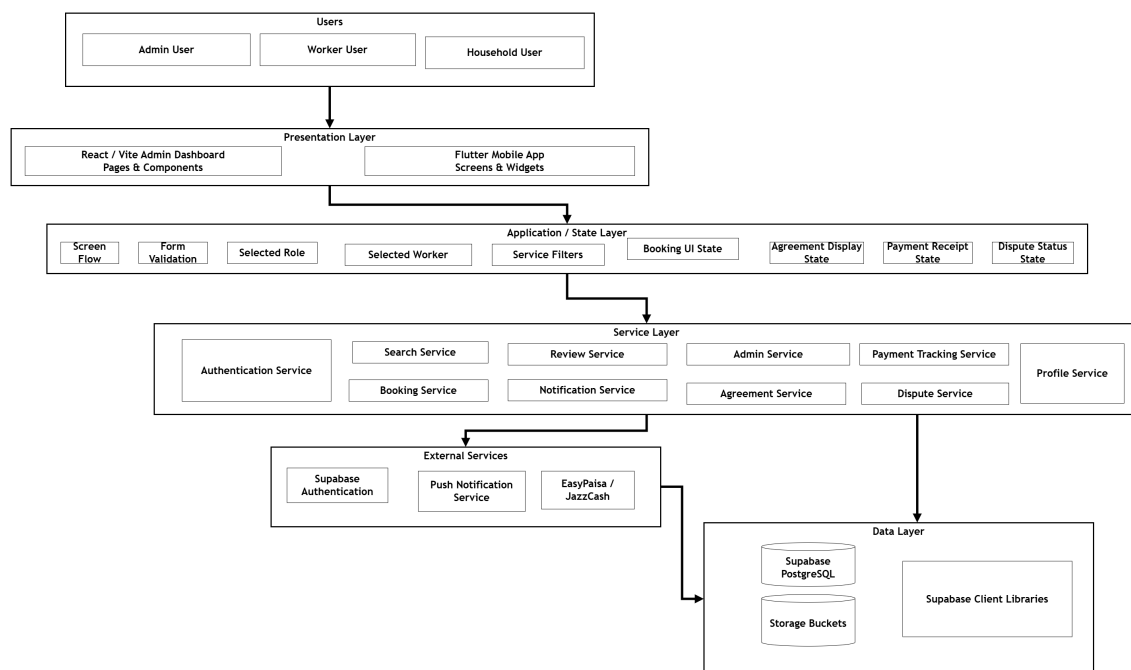


Figure 4.1 High-Level System Architecture of the HomeEase Platform

4.2 Process Flow Representation

The platform orchestrates multi-step, multi-actor operational workflows across the booking, marketplace, recommendation, and verification cycles:

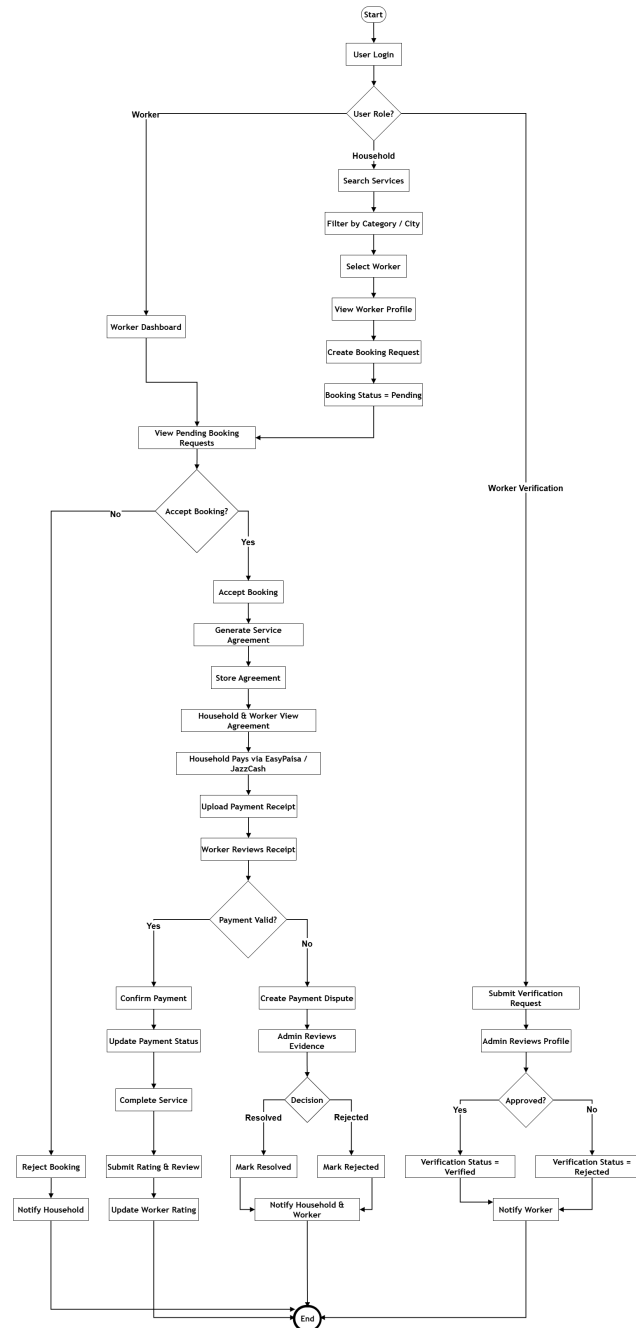


Figure 4.2 End-to-End Operational Process Flow in HomeEase

4.3 Design Models

4.3.1 Class Diagram

The object-oriented design model encapsulates the system domain into cohesive entity and service classes:

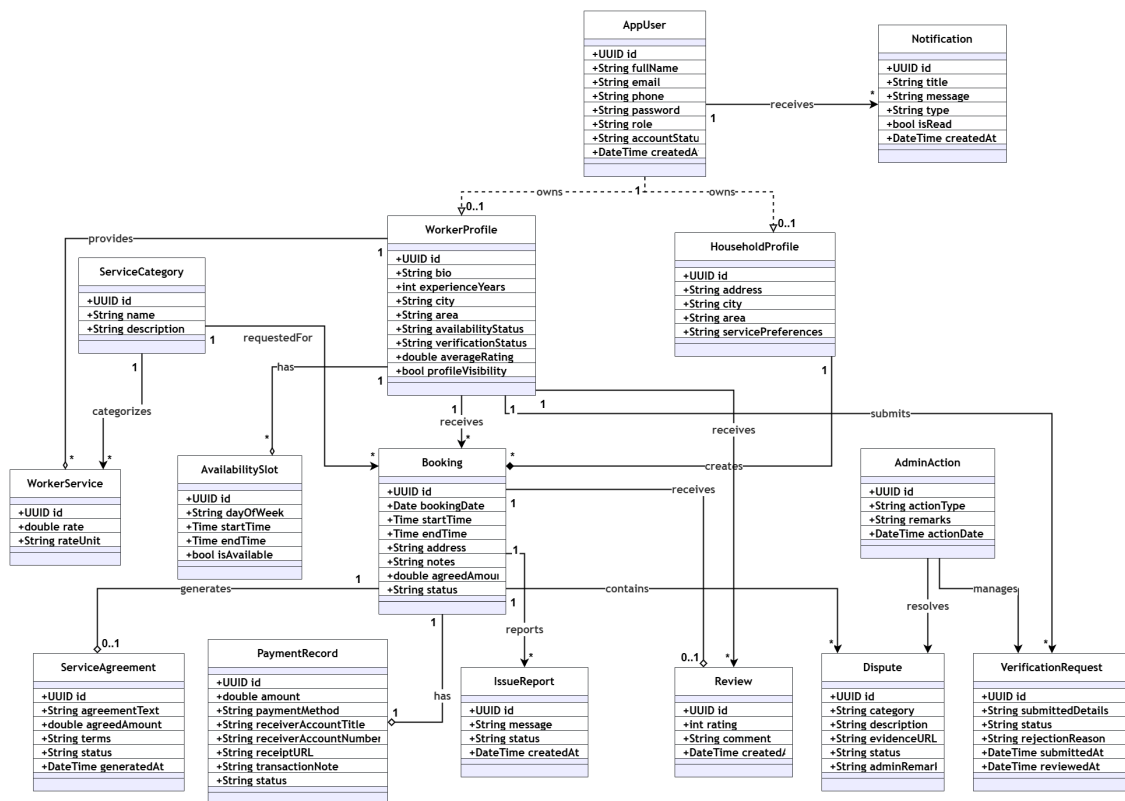


Figure 4.3 Class Diagram of HomeEase Core Domain Models

4.3.2 Sequence Diagram

The sequence model formalizes dynamic message passing between user interfaces, controllers, recommendation engines, and database services during the AI Search, Booking Request, and Job Application lifecycles.

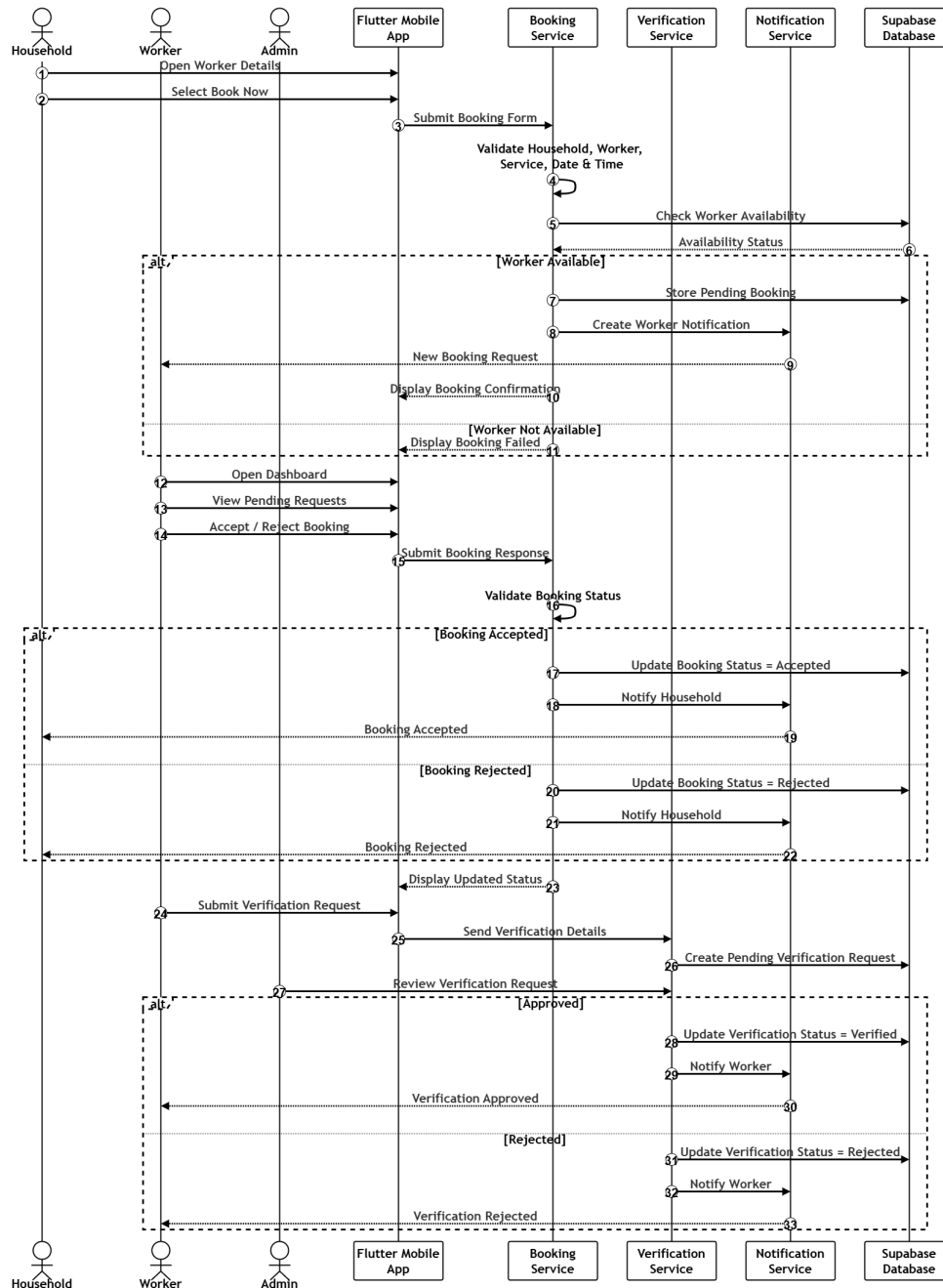


Figure 4.4 Sequence Diagram for AI Recommendation and Booking Lifecycle

4.3.3 State Transition Model

HomeEase coordinates several critical state machines:

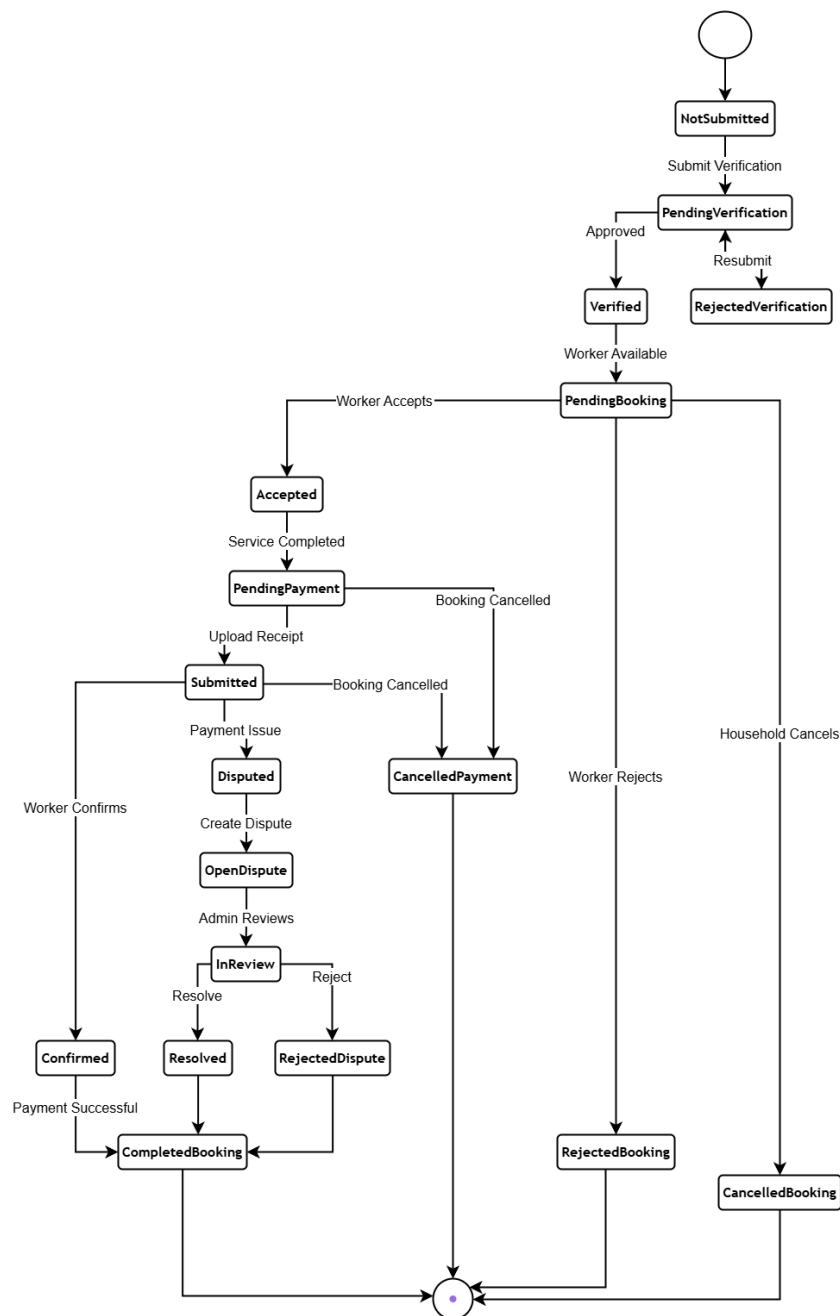


Figure 4.5 State Transition Diagram for Booking, Job Post, and Verification Lifecycles

4.3.4 Entity-Relationship Diagram & Data Dictionary

The relational database schema is modeled in PostgreSQL and normalized to Third Normal Form (3NF). It comprises 16 relational tables:

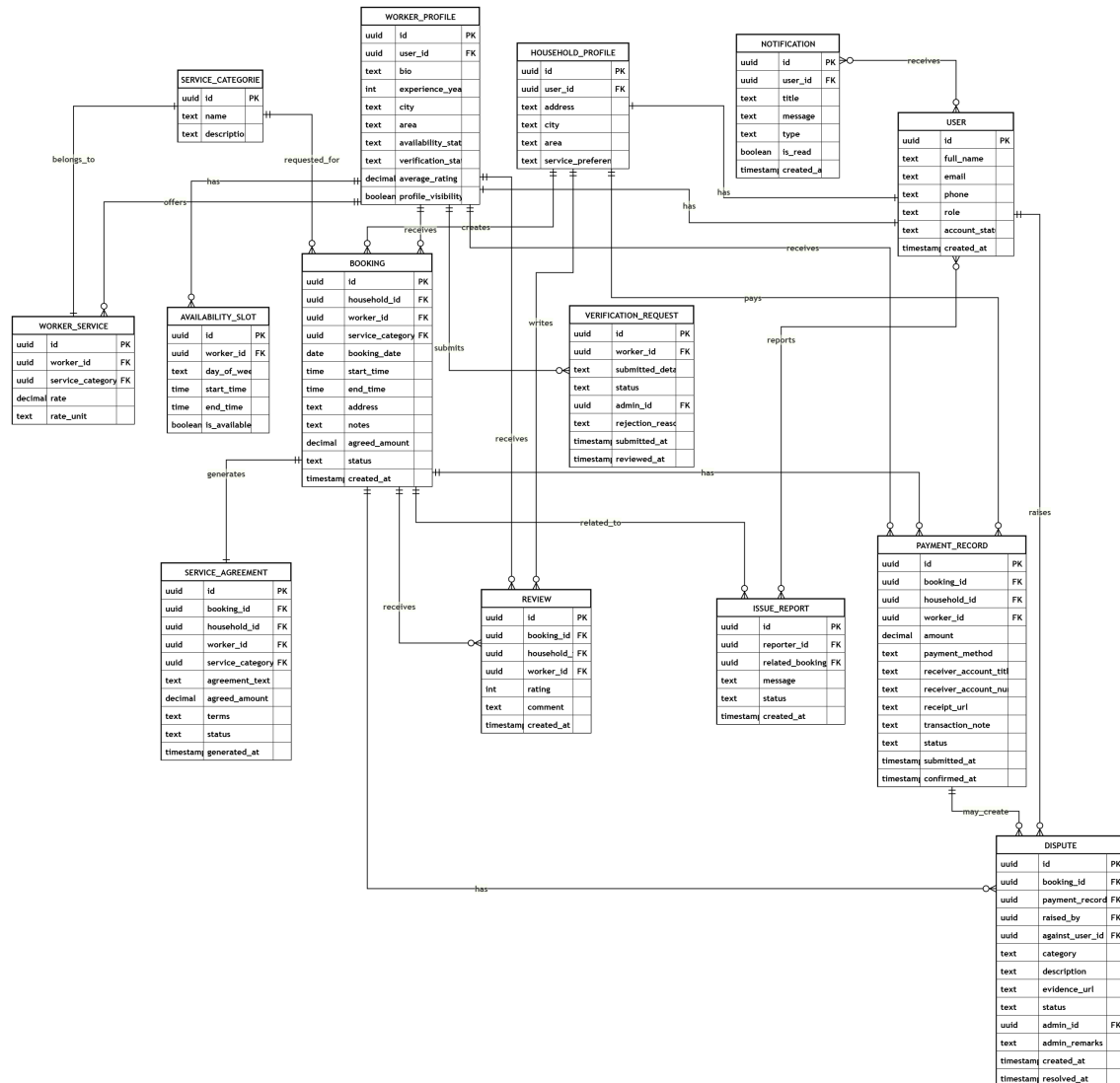


Figure 4.6 Entity-Relationship Diagram of the HomeEase PostgreSQL Database

Data Dictionary Table Overview:

Table 4.1 HomeEase Database Data Dictionary

Table Name	Primary Key	Foreign Keys	Functional Purpose
USERS	id (UUID)	None	Master account records, email, phone, role, password hash, account status.
HOUSEHOLD_PROFILES	id (UUID)	user_id -> USERS(id)	Household employer profile, Abbottabad address, locality, preferred language.

Table 4.1 HomeEase Database Data Dictionary (continued)

Table Name	Primary Key	Foreign Keys	Functional Purpose
WORKER_PROFILES	id (UUID)	user_id -> USERS(id)	Worker profile, experience years, bio, latitude, longitude, verified flag, rating.
SERVICE_CATEGORIES	id (UUID)	None	Service classifications (Cooking, Cleaning, Childcare, Elderly Care, Maid).
WORKER_SERVICES	id (UUID)	worker_id, category_id	Associative table mapping worker specializations with hourly/visit rates.
AVAILABILITY_SLOTS	id (UUID)	worker_id -> WORKER_PROFILES(id)	Weekly day-of-week worker availability intervals and active toggles.
BOOKINGS	id (UUID)	household_id, worker_id, category_id	Central transactional entity linking employer, worker, date, hours, and status.
SERVICE_AGREEMENTS	id (UUID)	booking_id, household_id, worker_id	Auto-generated formal service contract text, agreed fee, and signing status.
JOB_POSTS	id (UUID)	household_id, category_id	Bidirectional marketplace open gig listings with budget, locality, and date.
JOB_APPLICATIONS	id (UUID)	job_post_id, worker_id	Worker applications submitted against open gigs with proposed rates.
PAYMENT_RECORDS	id (UUID)	booking_id, household_id, worker_id	Manual payment audit records (Cash/JazzCash/EasyPaisa), receipt URLs, confirmation.
REVIEWS	id (UUID)	booking_id, household_id, worker_id	Multi-factor ratings (punctuality, behavior, quality) and written comments.
VERIFICATION_REQUESTS	id (UUID)	worker_id, admin_id	KYC audit submissions containing CNIC front/back URLs and review remarks.
DISPUTES	id (UUID)	booking_id, raised_by, admin_id	Transaction grievances, evidence URLs, mediation status, and admin resolution.
NOTIFICATIONS	id (UUID)	user_id -> USERS(id)	In-app and push notification alerts for booking status and gig updates.
ISSUE_REPORTS	id (UUID)	reporter_id, booking_id	General platform feedback, technical bug reports, and safety alerts.

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